14 JUNE 1965

SUPERSEDING
MIL-W-13276(SigC)
1 March 1954

MILITARY SPECIFICATION

WIRE, ELECTRICAL; ANTENNA, (WIRE W-28)

1. SCOPE

1.1 Wire W-28 consists of 7 strands of No. 20 AWG silicon bronze wire concentric stranded, forming a tight cylindrical conductor and is suitable for antenna applications.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-12000 Cable, Cord, and Wire, Electric; Packaging of

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Table for Inspection by Attributes.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The material for the wire shall be as specified herein. The best material commercially available for the purpose shall be used when a definite material is not designated.

FSC 6145

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3.2 Strands.- Each strand of W-28 wire shall consist of silicon bronze alloy wire having a diameter of 0.032 inch ±0.001 inch (No. 20 AWG.)

3.3 Conductor construction.- Wire W-28 shall be constructed of 7 strands of wire as specified in 3.2, concentrically stranded with a left hand lay to form a tight cylindrical conductor. The length of lay shall be between 7.5 inches and 15 inches.

3.4 Joints in conductors.- Factory joints in conductors shall be avoided, and in no case shall they exceed two joints per strand per length of wire, with joints spaced not less than 50 feet apart. All joints shall be butt brazed using silver alloy solder and a nonacid flux, and shall be finished smoothly in a workmanlike manner. No twist joints shall be used. The tensile strength of a section of each strand, which includes the joint, shall be not less than 90 percent of the tensile strength of an adjacent section of the strand of equal length without the joint.

3.5 Breaking strength.- Wire W-28 shall have a breaking strength of not less than 600 pounds.

3.6 Elongation.- The elongation of wire W-28 shall be not less than 0.80 percent of the zero length of stranded conductor. The elongation shall be determined by measurements made between the jaws of the testing machine. The zero length shall be the distance between the jaws when a load equal to 60 lbs is applied. The zero length shall be as near 60 inches as practicable. The final length shall be the distance between the jaws at the time of rupture.

3.7 D.C. resistance.- The d.c. resistance of wire W-28 shall not exceed 10.5 ohms per 1000 feet at or corrected to 68°F.

3.8 Workmanship.- Strands and completed Wire W-28, shall be manufactured and finished in a thoroughly workmanlike manner.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

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FOLD

HEADQUARTERS
U. S. ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

Commanding Officer
U. S. Army Electronics Command
Fort Monmouth, New Jersey 07703

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GPO 201-911

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD FORM 1426 1 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

Army-Ft Monmouth, NJ-MON 94-64

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